

Loops in Java

1. For Loop

- A **for loop** is a simpler way to handle loops with:
 - **Initialization**
 - **Condition**
 - **Update**

While Loop Equivalent

```
int i = 0;                // initialization
while (i < 5) {           // condition
    System.out.println("i = " + i);
    i++;                 // update
}
```

Simplified For Loop

```
for (int i = 0; i < 5; i++) {
    System.out.println("i = " + i);
}
```

Why Use a For Loop?

1. **Easy to read and shorter** → all in one line
2. **Prevent mistakes** → less chance of forgetting the update (i++)
3. **Intent is clear**
 - `for` → repeat a fixed number of times
 - `while` → repeat until a condition becomes false (not necessarily a fixed count)

multiple statement

In a for loop, when you have more than one statement in the initialization or update sections, you separate them with commas.

Important points:

The **condition cannot have commas**, it must be a single boolean expression.

```
for (int x = 1, y = 5; x <= 3 && y >= 3; x++, y--) {
    System.out.println("x = " + x + ", y = " + y);
}
```

output:

```
x = 1, y = 5
x = 2, y = 4
x = 3, y = 3
```
